2.70 2D Matrices: Transformations

Worked Solutions

1 a)

Stretch parallel to the x-axis scale factor 2 and stretch parallel to the y-axis scale factor 3

1 b)

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2-\lambda & 0 \\ 0 & 3-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (2-\lambda)(3-\lambda)$$

So $(2-\lambda)(3-\lambda) = 0$

$$\therefore \lambda = 2 \text{ or } \lambda = 3$$

When $\lambda_1 = 2$ then $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow y = 0 \text{ for any value of } x \text{ so } \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

When $\lambda_2 = 3$ then $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}$

$$\begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -x = 0 \text{ for any value of } y \text{ so } \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\therefore \lambda_1 = 2, \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \lambda_2 = 3, \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

1 c)

$y = 0$ and $x = 0$

1 d)

$\det \mathbf{A} = 6$ so the area of the image has been enlargened by a factor of 6

2 a)

Reflection across the line $y = x$

2 b)

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0-\lambda & 1 \\ 1 & 0-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (-\lambda)(-\lambda) - 1$$

So $\lambda^2 - 1 = 0$

$$\therefore \lambda = -1 \text{ or } \lambda = 1$$

When $\lambda_1 = -1$ then $\mathbf{A} + \mathbf{I} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow x = -y \text{ so } \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

When $\lambda_2 = 1$ then $\mathbf{A} - 1\mathbf{I} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -x + y = 0 \text{ so } \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\therefore \lambda_1 = -1, \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ and } \lambda_2 = 1, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

2 c)

$y = -x$ and $y = x$

2 d)

$\det \mathbf{A} = -1$ so the area is preserved, but the orientation is reversed.

3 a)

Rotation 90° anti-clockwise about the origin.

3 b)

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 - \lambda & -1 \\ 1 & 0 - \lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (-\lambda)(-\lambda) + 1$$

So $\lambda^2 + 1 = 0$

$$\therefore \lambda = -i \text{ or } \lambda = i$$

3 c)

Since the eigenvalues are complex, there are no invariant lines.

Geometrically, since every line rotates, no real line maps onto itself.

3 d)

$\det \mathbf{A} = 1$ so the area and orientation is preserved.

4 a)

$$\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2-\lambda & 0 \\ 0 & -2-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (2-\lambda)(-2-\lambda)$$

So $(2-\lambda)(-2-\lambda) = 0$

$$\therefore \lambda = -2 \text{ or } \lambda = 2$$

When $\lambda_1 = -2$ then $\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}$

$$\begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow 4x = 0 \text{ so } \mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

When $\lambda_2 = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 0 & 0 \\ 0 & -4 \end{pmatrix}$

$$\begin{pmatrix} 0 & 0 \\ 0 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -4y = 0 \text{ so } \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\therefore \lambda_1 = -2, \mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ and } \lambda_2 = 2, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

4 b)

$y = 0$ and $x = 0$

4 c)

$\det \mathbf{A} = -4$ so the area is enlarged by factor of 4, and the orientation is reversed.

4 d)

The eigenvalues and eigenvectors tell us:

- A stretch of scale factor 2 in the direction $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$
- A stretch of scale factor 2 in the direction $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ with a reflection in the x-axis

$\therefore$ Enlargement of scale factor 2 and a reflection in the x-axis.

5 a)

$$\begin{pmatrix} \frac{13}{5} & \frac{6}{5} \\ \frac{6}{5} & \frac{22}{5} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} \frac{13}{5} & \frac{6}{5} \\ \frac{6}{5} & \frac{22}{5} \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{13}{5}-\lambda & \frac{6}{5} \\ \frac{6}{5} & \frac{22}{5}-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (\frac{13}{5} - \lambda)(\frac{22}{5} - \lambda) - \frac{36}{25}$$

So $\lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5) = 0$

$$\therefore \lambda = 2 \text{ or } \lambda = 5$$

5 b)

When $\lambda_1 = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} \frac{3}{5} & \frac{6}{5} \\ \frac{6}{5} & \frac{12}{5} \end{pmatrix}$

$$\begin{pmatrix} \frac{3}{5} & \frac{6}{5} \\ \frac{6}{5} & \frac{12}{5} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow \frac{3}{5}x + \frac{6}{5}y = 0 \text{ so } y = -\frac{1}{2}x$$

When $\lambda_2 = 5$ then $\mathbf{A} - 5\mathbf{I} = \begin{pmatrix} -\frac{12}{5} & \frac{6}{5} \\ \frac{6}{5} & -\frac{3}{5} \end{pmatrix}$

$$\begin{pmatrix} -\frac{12}{5} & \frac{6}{5} \\ \frac{6}{5} & -\frac{3}{5} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -\frac{12}{5}x + \frac{6}{5}y = 0 \text{ so } y = 2x$$

$$\therefore y = -\frac{1}{2}x \text{ and } y = 2x$$

5 c)

The eigenvalues and eigenvectors tell us:

- A stretch of scale factor 2 in the direction $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$
- A stretch of scale factor 5 in the direction $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

They are perpendicular so there are no rotations or shears.

$\therefore$ A stretch of scale factor 2 along $y = -\frac{1}{2}x$ and a stretch of scale factor 5 along $y = 2x$

6 a)

Since $2\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$ then $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ is an eigenvector.

6 b)

$\lambda = 2$

7 a)

Since $-2\begin{pmatrix} 2 \\ -5 \end{pmatrix} = \begin{pmatrix} -4 \\ 10 \end{pmatrix}$ then $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$ is an eigenvector.

7 b)

$\lambda = -2$

8 a)

$$\begin{pmatrix} p & q \\ q & p \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} p & q \\ q & p \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p-\lambda & q \\ q & p-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (p-\lambda)(p-\lambda) - q^2$$

So $\lambda^2 - 2\lambda p + p^2 - q^2 = 0$

$$(\lambda - (p-q))(\lambda - (p+q)) = 0$$

$$\therefore \lambda = p - q \text{ or } \lambda = p + q$$

When $\lambda_1 = p - q$ then $\mathbf{A} - (p-q)\mathbf{I} = \begin{pmatrix} q & q \\ q & q \end{pmatrix}$

$$\begin{pmatrix} q & q \\ q & q \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow qx + qy = 0 \text{ so } \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

When $\lambda_2 = p + q$ then $\mathbf{A} - (p+q)\mathbf{I} = \begin{pmatrix} -q & q \\ q & -q \end{pmatrix}$

$$\begin{pmatrix} -q & q \\ q & -q \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -qx + qy = 0 \text{ so } \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\therefore \lambda_1 = p - q, \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ and } \lambda_2 = p + q, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

8 b)

Invariant lines are $y = -x$ and $y = x$

Consider gradients: $-1 \times 1 = -1$ so the invariant lines are perpendicular.

8 c)

The eigenvalues and eigenvectors tell us:

- A stretch of scale factor $p - q$ in the direction $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$
- A stretch of scale factor $p + q$ in the direction $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

They are perpendicular so there are no rotations or shears.

$\therefore$ A stretch of scale factor $p - q$ along $y = -x$ and a stretch of scale factor $p + q$ along $y = x$

9 a)

$$\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (3-\lambda)(3-\lambda) - 1$$

So $\lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2) = 0$

$$\therefore \lambda = 2 \text{ or } \lambda = 4$$

9 b)

When $\lambda_1 = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow x + y = 0 \text{ so } y = -x$$

When $\lambda_2 = 4$ then $\mathbf{A} - 4\mathbf{I} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -x + y = 0 \text{ so } y = x$$

$$\therefore y = -x \text{ and } y = x$$

9 c)

$\mathbf{A}^2$ has eigenvalues 4 and 16

- A stretch of scale factor 4 in the direction $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$
- A stretch of scale factor 16 in the direction $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

They are perpendicular so there are no rotations or shears.

So $\mathbf{A}^2$ stretches along the line $y = -x$ by scale factor 4 and along the line $y = x$ by scale factor 16

10)

$$\begin{pmatrix} -10 & 8 \\ -16 & 14 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} -10 & 8 \\ -16 & 14 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -10-\lambda & 8 \\ -16 & 14-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (-10-\lambda)(14-\lambda) + 128$$

So $\lambda^2 - 4\lambda - 12 = (\lambda + 2)(\lambda - 6) = 0$

$$\therefore \lambda = -2 \text{ or } \lambda = 6$$

When $\lambda_1 = -2$ then $\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} -8 & 8 \\ -16 & 16 \end{pmatrix}$

$$\begin{pmatrix} -8 & 8 \\ -16 & 16 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -8x + 8y = 0 \text{ so } x = y$$

When $\lambda_2 = 6$ then $\mathbf{A} - 6\mathbf{I} = \begin{pmatrix} -16 & 8 \\ -16 & 8 \end{pmatrix}$

$$\begin{pmatrix} -16 & 8 \\ -16 & 8 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -16x + 8y = 0 \text{ so } y = 2x$$

$$\therefore y = x \text{ and } y = 2x$$

$\mathbf{A}^3$ has eigenvalues $(-2)^3 = -8$ and $6^3 = 216$

- A stretch of scale factor -8 in the direction $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$
- A stretch of scale factor 216 in the direction $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

So $\mathbf{A}^3$ stretches along the line $y = x$ by scale factor 8, with a change in orientation, and along the line $y = 2x$ by scale factor 216 with orientation preserved.